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Machine-Learning Model Design & Implementation

NASA Space Apps Challenge

1. Final ML Architecture

NASA PSI DATA

```
+-- FLEX -----> Liquid-fire extinction classifier
+-- BASS-II ----> Solid-fire classifier + spread regressor
+-- SAFFIRE-I --> Spacecraft-material burn-time regressor
+-- SAME -----> Smoke/detector model after extracting a real response target
+-- SPICE -----> Smoke-point / flame-length regressor
```

MODEL ROUTER -> DOMAIN CHECK -> PREDICTION -> UNCERTAINTY -> EXPLANATION -> NASA TRACEABILITY -> DASHBOARD

Module	NASA source	Problem	Output
Liquid Fire	FLEX (PSI-69)	Classification	P(extinction), P(continued burning)
Solid Fire	BASS-II (PSI-25)	Classification + regression	Outcome and/or flame-spread rate
Large-scale Fire	SAFFIRE-I (PSI-98)	Regression	Burn duration
Smoke Detection	SAME (PSI-102)	After target extraction	Detector or particle response
Soot/Smoke Point	SPICE (PSI-107)	Regression	Smoke point / flame length

2. Official NASA Datasets

FLEX - PSI-69

Liquid-fuel droplet flammability/extinction; pressure, oxygen, suppressant conditions and combustion measurements. Best use: liquid-fire extinction classifier.

Official NASA source: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-69>

BASS-II - PSI-25

Solid-fuel ignition, flame growth, spread and extinction. Primary variables include flow velocity/direction, oxygen concentration and sample geometry.

Official NASA source: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-25>

SAFFIRE-I - PSI-98

Experimental table includes material, sample dimensions, airflow, O₂, ignition power/time, burn time and flow direction. Best use: burn-time regression.

Official NASA source: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-98>

SAME - PSI-102

Visible conditions include sample material, airflow, weight-loss rate and aging duration. Extract a genuine detector/particle response before modelling smoke response.

Official NASA source: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-102>

SPICE - PSI-107

Fuel/burner/co-flow conditions plus smoke-point and flame measurements. Best use: smoke-point/flame-length regression.

3. CSV Files to Create

flex_clean.csv

Field	Role	Meaning
experiment_id	ID	Traceability only
fuel_material	X	Fuel
oxygen_pct	X	O2 %
pressure_kpa	X	Pressure
droplet_size_mm	X	Droplet size
suppressant_type	X	Suppressant
suppressant_pct	X	Concentration
temperature_c	Optional X	If consistently measured
outcome	y	Source-derived outcome
source_ref	Audit	NASA source

bassii_clean.csv

Field	Role
experiment_id	ID
fuel_material	X
oxygen_pct	X
airflow_cm_s	X
flow_direction	X
geometry	X
flame_spread_rate	y regression, if available
outcome	y classification, if available
source_ref	Audit

saffire1_clean.csv

Field	Role
sample_number	ID
material	X
sample_thickness	X
sample_length	X
sample_width	X
airflow	X
oxygen_pct	X
ignition_power_w	X
ignition_time_s	X
flow_direction	X
burn_time_s	y
source_ref	Audit

same_clean.csv

Field	Role
sample_id	ID
sample_material	X
airflow_cm_s	X
weight_loss_rate_mg_min	X
aging_duration_min	X
detector_response_or_particle_metric	y - extract from analyzed NASA data
source_ref	Audit

Do not invent a smoke-danger target if NASA data do not define one.

spice_clean.csv

Field	Role
-------	------

experiment_id	ID
fuel_material	X
burner_diameter_mm	X
coflow_velocity_cm_s	X
fuel_flow_rate	X
smoke_point	y option 1
flame_length	y option 2
radiative_heat_loss	optional
source_ref	Audit

4. Model A - FLEX Liquid-Fire Classifier

Input: fuel + O2 + pressure + droplet size + suppressant conditions. Output: source-defined flame-outcome probabilities.

Illustrative input (not NASA measurement)

```
fuel_material = methanol
oxygen_pct = 17.0
pressure_kpa = 85.0
droplet_size_mm = 3.0
suppressant_type = CO2
suppressant_pct = 5.0
```

Illustrative output

```
predicted_class = EXTINGUISHED
P(extinguished) = 0.81
P(continued_burning) = 0.19
confidence_band = HIGH
in_domain = true
```

Model	Purpose
Logistic Regression	Transparent baseline
Random Forest	Nonlinear mixed tabular data
Gradient Boosting / XGBoost	Strong tabular comparison model

```
features =
['fuel_material', 'oxygen_pct', 'pressure_kpa', 'droplet_size_mm', 'suppressant_type', 'suppressant_pct']
target = 'outcome'
```

```
# numeric: median imputation + scaling
# categorical: most-frequent imputation + one-hot encoding
# validate with StratifiedKFold / GroupKFold
# report precision, recall, F1, confusion matrix, ROC-AUC and calibration
```

5. Model B - BASS-II

Outcome classifier

Input: material + oxygen_pct + airflow_cm_s + flow_direction + geometry

Illustrative output: P(extinction)=0.67; P(continued_burning)=0.33

Flame-spread regressor

Input: material + oxygen_pct + airflow_cm_s + flow_direction + geometry

Illustrative output: predicted_flame_spread_rate = 2.1 mm/s

Only show an interval after implementing a validated uncertainty method.

Recommended: Random Forest and Gradient Boosting/XGBoost, compared with a simple baseline. Regression metrics: MAE, RMSE, R2 and residual plots.

6. Model C - SAFFIRE-I Burn-Time Regressor

```
X = [material, sample_thickness, sample_length, sample_width, airflow, oxygen_pct, ignition_power_w,
ignition_time_s, flow_direction]
y = burn_time_s
```

Illustrative output:

```
predicted_burn_time = 46.2 s
```

```
in_domain = true
```

```
uncertainty_range = only if generated by validated conformal/bootstrap method
```

Compare Ridge Regression, Random Forest Regressor and Gradient Boosting/XGBoost Regressor.

7. Model D - SAME Smoke/Detector Model

Use sample material, airflow, weight-loss rate and aging duration as condition features only after extracting a genuine detector-response or particle-response target from NASA analyzed data.

```
If y is numeric -> regression
```

```
If NASA provides a documented categorical response -> classification
```

```
Do not create an unsupported high/low smoke label.
```

8. Model E - SPICE

Model E1:

```
X = fuel_material + burner_diameter_mm + coflow_velocity_cm_s + fuel_flow_rate
```

```
y = smoke_point
```

Model E2:

```
X = same feature family
```

```
y = flame_length
```

Compare a regularized linear baseline, Random Forest Regressor and Gradient Boosting/XGBoost. Report MAE, RMSE, R2 and residual behaviour.

9. Preprocessing and Data Quality

Check	Action
Units	Convert each physical quantity to one documented canonical unit
Missing values	Inspect why values are missing; impute only when reasonable
Categorical fields	One-hot encode with unknown-category handling
Identifiers	Never use IDs/filenames as predictive features
Duplicates	Remove accidental duplicates, retain legitimate repeated experiments
Leakage	Group related runs during splitting where possible
Source tracking	Retain source_ref and raw NASA package
Unknown categories	Do not silently map an unseen material to a known one

10. Validation

Task	Metrics
Classification	Precision, recall, F1, confusion matrix, ROC-AUC where applicable, calibration/Brier score
Regression	MAE, RMSE, R2, residual plot, predicted-vs-actual
Probability	Calibration curve and Brier score
Generalization	GroupKFold or experiment-level holdout when possible

11. Confidence, Uncertainty and Domain Checks

Model probability is not the same as scientific certainty. Confidence should combine calibration, domain coverage and nearby evidence.

UI field	Meaning
Model probability	Calibrated class probability
Confidence band	Project-defined display label based on validated rules
Evidence density	Nearby training-data support
In-domain	Whether inputs fall within validated ranges/categories
Uncertainty range	Regression interval from a validated uncertainty method

Example rule after calibration: High may require calibrated probability ≥ 0.80 , in-domain input and adequate nearby evidence. Moderate covers intermediate support. Limited appears for sparse or borderline evidence. These are project rules, not NASA standards.

Out-of-domain example:

Input O2 = 35%

Validated training domain = 12% to 25%

Dashboard: Prediction withheld

Reason: O2 concentration is outside validated training range.

12. Explainability

Use SHAP or carefully presented feature importance for tree-based models. Label this as model explanation, not causal importance.

Illustrative explanation:

O2 concentration 38%

Pressure 24%

Suppressant level 18%

Droplet size 12%

Fuel identity 8%

13. Unified API

POST /analyze

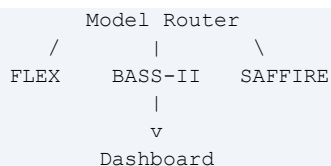
```
{
  "scenario_type": "liquid",
  "fuel_material": "methanol",
  "oxygen_pct": 17.0,
  "pressure_kpa": 85.0,
  "droplet_size_mm": 3.0,
  "suppressant_type": "CO2",
  "suppressant_pct": 5.0
}
```

Response:

```
{
  "model": "FLEX-v1.0",
  "prediction": "EXTINGUISHED",
  "probabilities": {"extinguished": 0.81, "continued_burning": 0.19},
  "in_domain": true,
  "confidence_band": "HIGH"
}
```

14. Sensor Integration

ESP32 sensors: O2 + Pressure + Airflow + Temperature
+ user-selected fuel/material/geometry
|
v



Only pass a sensor value into a model when the model was trained with a scientifically compatible field and unit.

15. Final Dashboard

Panel	Content
Scenario	Fire type plus user/sensor inputs
Prediction	Class probability or regression estimate
Confidence & domain	Confidence band, in-domain flag, validated uncertainty
Why this result	SHAP/feature explanation
Model quality	Cross-validated F1/AUC or MAE/RMSE/R2
Training coverage	Input ranges/categories and evidence density
NASA provenance	Dataset, PSI ID/DOI/source reference
Caveats	Out-of-domain and scientific limitations

Illustrative result card

```

ASTRAFLAME FIRE-BEHAVIOUR ANALYSIS
Scenario: Liquid fuel
Prediction: Extinguished
Probability: 81%
Confidence: High (project rule; calibrated model)
Domain check: PASS
Top driver: O2 concentration
NASA training source: FLEX / PSI-69
Model quality: populate only after actual training
Caveat: decision-support prediction, not certification
  
```

16. Competition-Ready Folder Structure

```

AstraFlame-ML/
  data/raw/{flex,bass_ii,saffire,same,spice}/
  data/processed/{flex_clean.csv,bassii_clean.csv,saffire1_clean.csv,same_clean.csv,spice_clean.csv}
  preprocessing/
  training/
  models/
  explainability/
  api/
  reports/
  README.md
  
```

Store the untouched NASA downloads under data/raw. Store normalized experiment-level CSVs under data/processed. Save preprocessing and model together. Version models and retain metrics/model cards.

17. Recommended Execution Order

Action
Download FLEX and build flex_clean.csv; train and calibrate Model A
Download BASS-II; build classifier/regressor only for targets actually present
Use SAFFIRE-I table for burn-time regression
Extract SAME analyzed response before training smoke model
Train SPICE smoke-point/flame-length regression
Add domain checks, uncertainty, SHAP and model cards
Expose all validated models through one router/API and dashboard

18. NASA Source Links

FLEX: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-69>

BASS-II: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-25>

SAFFIRE-I: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-98>

SAME: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-102>

SPICE: <https://psi.nasa.gov/physci/repo/data/investigations/PSI-107>

NASA PSI Repository: <https://psi.nasa.gov/physci/repo>